
The Combinatorics Masterclass

Topic 1: Fundamental Principles of Counting

This is the absolute bedrock of all combinatorics. Almost every problem uses these ideas, either directly or indirectly.

Rule of Sum (The “OR” Rule): If you can do Task A in m ways and Task B in n ways, and you cannot do both tasks at the same time, then you can choose to do either Task A OR Task B in $m + n$ ways.

Rule of Product (The “AND” Rule): If a procedure can be broken down into two stages, and there are m outcomes for the first stage and n outcomes for the second stage (regardless of the outcome of the first), then the total procedure can be carried out in $m \times n$ ways. This is for when you do Task A AND Task B.

Principle of Complementary Counting: Sometimes, counting what you want is hard. It can be easier to count the total number of possibilities and subtract the number of possibilities you don't want. $(\text{Total Ways}) - (\text{Unwanted Ways}) = (\text{Wanted Ways})$.

Practice Problem

How many 4-digit numbers can be formed using the digits $\{0, 1, 2, 3, 4, 5\}$ if the number must be even and repetition of digits is not allowed?

(Hint: This is tricky because the last digit being 0 affects the choices for the first digit. Consider breaking it into two separate cases using the Rule of Sum.)

Topic 2: Permutations and Combinations

These are the two most famous tools in counting. The key is understanding the difference.

Permutations (Arrangements): Used when the order of selection matters. A permutation is an arrangement of objects. The number of ways to arrange r objects from a set of n unique objects is:

$$P(n, r) = \frac{n!}{(n - r)!}$$

Combinations (Selections): Used when the order of selection does not matter. A combination is a selection of objects. The number of ways to choose r objects from a set of n unique objects is:

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n - r)!}$$

Practice Problem

A school committee of 5 people is to be formed from a group of 8 teachers and 6 students. In how many ways can the committee be formed if it must contain at least 3 teachers?

(Hint: “At least 3” means it could be 3 teachers OR 4 teachers OR 5 teachers. Use combinations for selection and the rule of sum for the cases.)

Topic 3: The Pigeonhole Principle (PHP)

This is a simple-sounding but incredibly powerful idea that is a favorite in Olympiads. It can solve problems that seem to have too little information.

Basic Form: If you have $n + 1$ pigeons and you want to place them into n pigeonholes, then at least one pigeonhole must contain two or more pigeons.

The key skill: Identifying the “pigeons” (the objects you are distributing) and the “pigeonholes” (the categories you are sorting them into).

Practice Problem

Show that in any group of 13 people, at least two of them must have their birthday in the same month.

(Hint: What are the pigeons? What are the pigeonholes? This is a direct application of the principle.)

A more advanced practice problem

Suppose 51 points are placed randomly inside a square of side length 1 meter. Prove that it is always possible to find three of those points that can be covered by a circular disk of radius $1/7$ meter.

(Hint: Divide the 1×1 square into smaller, identical squares (the pigeonholes). How small must they be to guarantee the condition? A 5×5 grid of 25 squares works well.)

Topic 4: Principle of Inclusion-Exclusion (PIE)

This is the formal name for a technique used to count the size of the union of several sets, making sure no element is counted more than once. It's a more advanced version of complementary counting.

For two sets, A and B :

$$|A \cup B| = |A| + |B| - |A \cap B|$$

For three sets, A , B , and C :

$$|A \cup B \cup C| = |A| + |B| + |C| - (|A \cap B| + |A \cap C| + |B \cap C|) + |A \cap B \cap C|$$

The pattern is: (Sum of singles) - (Sum of pairs) + (Sum of triples) - ...

Practice Problem

How many integers from 1 to 1000 (inclusive) are divisible by 2, 3, or 5?

(Hint: Let A be the set of numbers divisible by 2, B be the set divisible by 3, and C be the set divisible by 5. Find $|A|$, $|B|$, $|C|$, $|A \cap B|$ (divisible by 6), $|A \cap C|$ (divisible by 10), $|B \cap C|$ (divisible by 15), and $|A \cap B \cap C|$ (divisible by 30). Then apply the PIE formula.)

Topic 5: Casework and Strategic Organization

(This is not a formula, but a crucial skill inspired by Zeitz's book.)

Many problems don't have a single, clean formula. **The solution is to break the problem down into smaller, manageable, non-overlapping cases.** The key is to be systematic so you don't miss a case or double-count. A good strategy is to fix one element or condition and build cases around it.

Practice Problem

How many integers between 100 and 999 have at least one digit that is a 7?

(Hint: A direct count is messy. You'd have to count numbers with one 7, two 7s, and three 7s. Instead, use complementary counting! Find the total number of 3-digit integers. Then, find the number of 3-digit integers that have NO 7s. Subtract the second from the first.)

Topic 6: “Stars and Bars” and Distributing Objects into Boxes

(A classic technique for a specific but common type of problem.)

When distributing objects into boxes, the method depends on whether the objects and boxes are **distinct** (distinguishable) or **identical** (indistinguishable). There are four fundamental cases:

Case 1: Identical objects into distinct boxes

(Classic “stars and bars”)

Formula: $\binom{n+k-1}{k-1}$

Example: Distribute 5 identical candies among 3 children.

Solution: $\binom{5+3-1}{3-1} = \binom{7}{2} = 21$ ways.

Case 2: Identical objects into identical boxes

(Partitions of n into at most k parts)

Formula: The number of ways is given by the **partition function** $p(n, k)$, which counts the number of ways to write n as a sum of at most k positive integers, disregarding order. There is no simple closed formula, but values can be found in tables or with generating functions.

Example: Distribute 4 identical balls into 2 identical boxes.

Solution: We want $p(4, 2)$, the number of ways to write 4 as a sum of at most 2 positive integers: $4 = 4 + 0$, $4 = 3 + 1$, $4 = 2 + 2$. So $p(4, 2) = 3$.

Case 3: Distinct objects into identical boxes

(Partitions of a set; counted by Bell numbers)

Formula: No simple formula; the number of ways is given by the Bell number B_n (number of partitions of an n -element set).

Example: Place 3 different books into identical boxes.

Solution: The possible groupings are: all together, all separate, or some together. There are 5 ways (the 3rd Bell number).

Case 4: Distinct objects into distinct boxes

Formula: k^n

Example: Place 3 different books into 2 different boxes.

Solution: Each book has 2 choices: $2^3 = 8$ ways.

Summary Table:

Objects	Boxes	Formula/Method	Notes
Identical	Distinct	$\binom{n+k-1}{k-1}$	Stars and Bars
Identical	Identical	Partition number	Integer partitions
Distinct	Identical	Bell number B_n	Set partitions
Distinct	Distinct	k^n	Each object independent

Practice Problem

You have 6 balls and 3 boxes. For each of the following cases, how many ways can you distribute the balls among the boxes?

- (a) The balls are identical, the boxes are distinct.
- (b) The balls are identical, the boxes are identical.
- (c) The balls are distinct, the boxes are distinct.
- (d) The balls are distinct, the boxes are identical.

Hint: Use stars and bars for (a), the partition function for (b), powers for (c), and the Bell number for (d).

Topic 7: Circular Permutations

This is a direct and important extension of linear permutations. It answers the question: “What if we arrange things in a circle instead of a line?”

When arranging objects in a line, ABC is different from BCA. But when arranging people around a circular table, if person A is to the left of B, who is to the left of C, this is the same arrangement as BCA and CAB (they are just rotations of each other).

The Rule: To arrange n distinct objects in a circle, we first “fix” one object’s position to eliminate rotational duplicates. Then, we arrange the remaining $n - 1$ objects in a line. The number of ways is $(n - 1)!$.

If we are choosing and arranging r objects in a circle from n distinct objects, the formula is:

$$Q(n, r) = \frac{P(n, r)}{r} = \frac{n!}{r(n - r)!}$$

Necklaces & Keyrings (Reflections): Sometimes, flipping the circle over doesn’t create a new arrangement (e.g., a necklace with beads). In these cases, you divide by 2: $(n - 1)!/2$ (for $n > 2$). For seating arrangements, reflections are usually considered different.

Practice Problem

King Arthur wishes to seat 6 knights (including Sir Lancelot and Sir Galahad) around his round table.

- In how many ways can he seat the 6 knights?
- In how many ways can he seat them if Sir Lancelot and Sir Galahad must sit next to each other?

(Hint for (b): Treat Lancelot and Galahad as a single “block”. Arrange this block and the other knights in a circle. Then, consider that within their block, Lancelot and Galahad can swap places.)

Topic 8: Permutations with Repetition (Multisets)

This handles the common situation where the items you are arranging are not all distinct.

How many ways can you arrange the letters of the word BOOK? If all letters were distinct (B, O₁, O₂, K), the answer would be 4!. But since the two “O”s are identical, BO₁O₂K is the same as BO₂O₁K. We have overcounted.

The Rule: To find the number of permutations of n objects where there are n_1 identical items of type 1, n_2 identical items of type 2, ..., and n_r identical items of type r , the formula is:

$$\frac{n!}{n_1!n_2! \cdots n_r!}$$

Practice Problem

Consider the word ARRANGEMENTS.

- (a) How many distinct arrangements of the letters in ARRANGEMENTS are there?
- (b) How many arrangements have all the vowels together?
- (c) How many arrangements have no two identical letters adjacent?
- (d) How many arrangements start and end with the same letter?
- (e) How many arrangements have all the N's separated by at least one other letter?

(Hint: Use the multinomial formula for (a), block/grouping strategies for (b) and (e), and inclusion-exclusion for (c) if needed.)

Topic 9: Derangements

This is a famous and fascinating type of permutation problem with a memorable story. It asks for arrangements where nothing ends up in its original spot.

A derangement is a permutation of the elements of a set, such that no element appears in its original position. The classic example is the “hat-check problem”: n people leave their hats at the door. When they leave, they each randomly grab a hat. In how many ways can it happen that no one gets their own hat back?

The Formula: The number of derangements of n objects, denoted D_n , is

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + \frac{(-1)^n}{n!} \right)$$

Short Derivation (using PIE): Let A_i be the set of arrangements where object i is in its original position. By PIE,

$$D_n = n! - \binom{n}{1}(n-1)! + \binom{n}{2}(n-2)! - \cdots + (-1)^n \binom{n}{n}(n-n)!$$

This simplifies to the formula above.

For small numbers, we can use casework and logic:

- $D_1 = 0$ (one person must get their own hat)
- $D_2 = 1$ (The only permutation is (1,2). The only derangement is (2,1).)
- $D_3 = 2$ (The permutations are (1,2,3), (1,3,2), (2,1,3), (2,3,1), (3,1,2), (3,2,1). The derangements are (2,3,1) and (3,1,2).)

Practice Problem

Four students (A, B, C, D) write their names on a piece of paper and put them in a box. The teacher randomly gives one paper back to each student. In how many ways can the papers be returned so that exactly one student receives their own paper?

(Hint: This is not a pure derangement problem, but it uses the same thinking. First, choose which student gets the correct paper. How many ways can you do that? Then, for the remaining three students, their papers must be a derangement. How many ways can you give the 3 papers to the 3 remaining students so that none of them get their own?)

Topic 10: Paths on a Grid (Lattice Paths)

This is a classic and highly visual problem type that beautifully connects to combinations.

These problems ask for the number of ways to travel from a starting point (like $(0,0)$) to an ending point (like (m,n)) on a grid, using only moves in specific directions (usually only Right and Up).

The Key Insight: To get from $(0,0)$ to (m,n) , you must make exactly m Right moves and n Up moves. The total journey has $m + n$ steps. The problem is simply choosing which of these $m + n$ steps will be the Right moves (the rest must be Up moves).

The Formula: The number of paths is $\binom{m+n}{m}$ or equivalently $\binom{m+n}{n}$.

Practice Problem

A spider is at the bottom-left corner of a 6×4 grid and wants to reach the top-right corner. It can only move Right or Up.

- How many different paths can the spider take?
- How many paths are there if the spider must pass through the point $(3,2)$?

(Hint for (b): Break the problem into two independent stages. First, count the paths from $(0,0)$ to $(3,2)$. Then, count the paths from $(3,2)$ to $(6,4)$. Use the Rule of Product.)

Topic 11: Double Counting (Counting in Two Ways)

This is not a formula, but a powerful proof technique. It establishes an identity or a property by showing that both sides of an equation are just two different ways of counting the same set of objects.

The strategy is as follows:

- Identify a quantity that can be counted.
- Method 1:** Count it in one way, which gives you an expression (Side A).
- Method 2:** Count the exact same quantity in a completely different way, which gives you another expression (Side B).

Since both methods counted the same thing, the results must be equal. Therefore, $A = B$. This is a very common way to prove combinatorial identities.

Famous Problem: The Handshake Lemma

At a party with n people, some people shake hands. If we add up the number of hands shaken by each person, the sum will be an even number.

Proof by Double Counting: Let's count the total number of handshakes that occurred.

Method 1: Let H be the total number of handshakes. This is our first count.

Method 2: Go to each person and ask them, “How many hands did you shake?” Let the answers be $d_1, d_2, \dots, d_n$. The sum of these answers is $d_1 + d_2 + \dots + d_n$. Now, consider a single handshake between Person X and Person Y. This one handshake was counted once by Person X and once by Person Y. So, every handshake is counted exactly twice in this sum. Therefore, $d_1 + d_2 + \dots + d_n = 2H$.

Conclusion: Since the sum equals $2H$, it must be an even number.

Practice Problem

Prove the “Committee Forming Identity”: $r \binom{n}{r} = n \binom{n-1}{r-1}$.

(Hint: Let’s count the number of ways to form a committee of r people from a group of n people, where one of the committee members is designated as the “Chairperson”.)

Method 1: First, choose the committee of r people. How many ways? Then, from those r people, choose one to be the Chairperson. How many ways? Use the Rule of Product.

Method 2: First, choose the Chairperson from the n people. How many ways? Then, from the remaining $n - 1$ people, choose the other $r - 1$ committee members. How many ways? Use the Rule of Product.)

Topic 12: Basic Probability

Many Olympiad problems frame counting questions in the language of probability. This is a direct and important application of combinatorics.

For events where all outcomes are equally likely, the probability of a specific event A happening is:

$$P(A) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Possible Outcomes}}$$

This means every basic probability problem is actually two combinatorics problems in one. You use all your counting skills (combinations, permutations, etc.) to find the numerator and the denominator.

Practice Problem

A committee of 3 people is to be randomly selected from a group of 5 men and 5 women. What is the probability that the committee will consist of exactly 2 women and 1 man?

(Hint: First, find the “Total Number of Possible Outcomes” by calculating how many ways you can choose any 3 people from the total group of 10. This is your denominator. Then, find the “Number of Favorable Outcomes” by calculating how many ways you can choose 2 women from 5 AND 1 man from 5. This is your numerator. Divide the two results.)